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Evaluation of physico-chemical characteristics of a fresh water pond of Samastipur, Bihar

Rahul Kumar[a] & Faiyaz Ahmad[b]*
[a]University Department of Zoology, B.R.A.Bihar University, Muzaffarpur, Bihar, India
[b]P.G. & UG Department of Zoology, L.S.College, Muzaffarpur, Bihar, India

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Abstract- The stepwise concentration of modern stable insecticides as they pass one step to another is hierarchy of animal kingdom show the phenomenon of biological magnification. Beside insecticides other important sources of pollution which disturb a pond ecosystem specially in Urban areas are human wastes. The aim of this study was to investigate physico-chemical quality of water of a fresh water pond of Samastipur district of Bihar and to detect the overall trophic status of ecosystem. Parameters for physico-chemical characteristics includes water temperature, transparency, pH, dissolved Oxygen (DO), alkalanity, Calcium, Magnesium, BOD, COD etc. The study comprises from July 2021 to June 2022.

Key words: Physico-chemical, Parameter, Samastipur, BOD, COD

INTRODUCTION

The structural components of an ecosystem consist of both biotic and abiotic components, the former comprises of producers, the only source of trapping solar energy converting it into chemical energy, the consumers of different strata and finally the decomposers. The various abiotic factors of physico-chemical like temperature, relative humidity, transparency, dissolved Oxygen (DO), Biological Oxygen Demand (BOD), chemical oxygen Demand (COD), pH, bicarbonate alkalinity, Magnesium, Chloride etc. constitute the abiotic factors. The use of different group of insecticides constitutes of Chlorinated hydrocarbon; though very effective act as a double edge sword, so far as pond ecology is concerned. Their action is twofold, due to their broad spectrum as they do extensive damage to pond eco-system. Their second drawback is the stable nature of their residue which accumulates in the system of those using the water or consuming the pond products like fishes, mollusks etc. often producing toxic hazard.

A large number of literatures is available on physico-chemical characteristic parameters from different parts of India, but literature is scanty about Samastipur pond. Srivastava and Kanungo (2013)[1] suggested that 70% of available water in India is polluted due to effluents from industries, waste, domestic sewage, agricultural drainage. Seasonal variation in the water quality is due to in the variation of population of specific algae. Kumar *et al.* (2017)[2] suggested that industrial wastes are being continuously added to the water bodies changes its water quality. The present study was aim to investigate whether the water quality of pond is use for drinking or fishing or not.

*Corresponding author :
Phone : 7258053166
E-mail : faiyaz266mjk@gmail.com

MATERIAL & METHODS

The physico-chemical characteristics analysis of a fresh water pond of Samastipur were done by the standard methods.[3,4] The water sample were collected between July 2021 to June 2022 at monthly interval. Some of the parameters were done at study site and for other characteristics brought to the laboratory for further analysis.

RESULT & DISCUSSION

Temperature is important for its effects on certain chemical and biological activities of organisms. The maximum temperature was recorded 30.0°C in June 2022 and minimum was 8.4°C in January 2022. Transparency was recorded maximum 35.2 cm in December 2021 whereas minimum 10.2 cm in August 2021. Conductivity was maximum 649 µmhos/cm in April 2022 whereas minimum 490 µmhos/cm in January 2022. Total dissolved solutes (TDS) are important parameter for detection of water quality Which is maximum 389 mg/l in August 2021 whereas minimum 305 mg/l in January 2022. The pH recorded maximum 8.5 in January 2022 whereas minimum 7 in July 2021. Dissolved Oxygen (DO) was maximum 10.4 mg/l in January 2022 whereas recorded minimum 5.02 mg/l in May 2022. Free carbonate was maximum 11.8 mg/l in September 2021 and 8.3 mg/l was minimum in August 2021. The maximum and minimum carbonate alkalinity was 22.2 mg/l and 11.7 mg/l in December 2021 and July 2021 respectively. The bicarbonate alkalinity was maximum 292.5 mg/l in March 2022 and minimum 162.5 mg/l in Dec. 2021. The calcium was maximum 26.2 mg/l in February 2022 whereas 14.3 mg/l was minimum in July 2021. Magnesium was recorded maximum 20.4 mg/l in January 2022 and minimum 8 mg/l in July 2021. Sulphate Sulphur maximum 30.8 mg/l in May 2022 and minimum 6.3 mg/l in September 2021. Phosphate and Phosphorus was maximum 0.827 mg/l in May 2022 whereas recorded minimum 0.483 mg/l in August 2021. Nitrate and Nitrogen was maximum 0.693 mg/l in March 2022 and minimum 0.513 mg/l in July 2021. The maximum amount of Chloride was 113.6 mg/l in April 2022 and minimum 49.8 mg/l in August 2021. BOD was recorded maximum and minimum as 5.9 mg/l in September 2021 and 2.5 mg/l in February 2022 respectively COD was recorded maximum 28.8 mg/l in May 2022 and minimum 14.0 mg/l in August 2021.

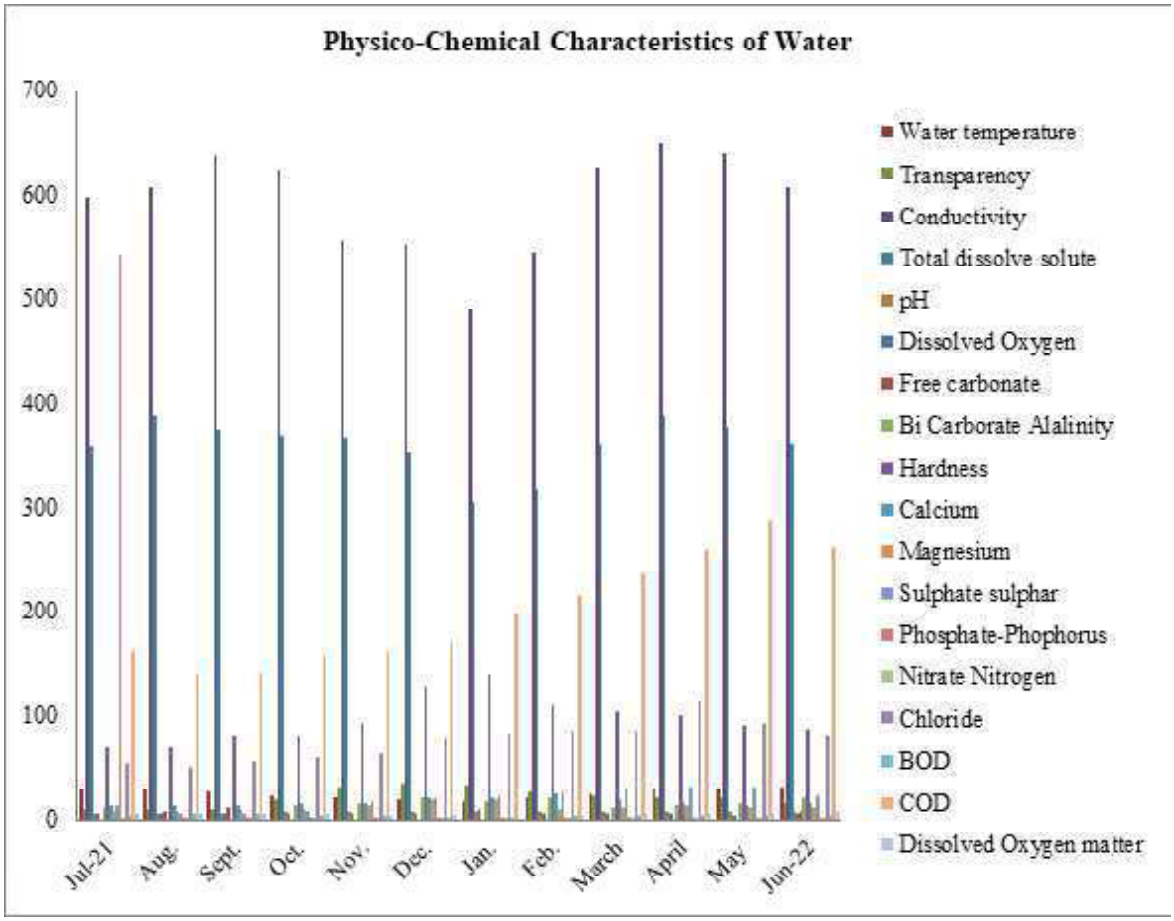
The present study reveal that the temperature was gradually increases from January 2022 to June 2022. The high temperature in summer a may be due to hot and long day. The Total Dissolved Solutes (TDS) has two-fold function, it affects the water transparency which in turn effect the photosynthesis and parts of the TDS supplies nutrients to the pond biotics. Due to the presence of the TDS the conductivity of water is increased. The CO_2 received during photosynthesis causes precipitation of $CaCO_3$ by algae preset in the water causing increase of water pH photosynthesis which alters the Dissolved Oxygen (DO), other physico-chemical and biological

Table 1: Physico-Chemical Characteristics of Water

Parameter/Months	July 2021	Aug. 2021	Sept. 2021	Oct. 2021	Nov. 2021	Dec. 2021	Jan. 2022	Feb. 2022	March 2022	April 2022	May 2022	June 2022
Water temperature	28.8	28.3	27.1	23	22.5	20.1	18.4	20.8	26.2	28.4	30	30.3
Transparency	10.8	10.2	10.4	18.8	30.5	35.2	32.8	27.3	24.4	21.8	21.4	15
Conductivity	598	608	638	625	556	552	490	544	626	649	640	608
Total dissolve solute	358	389	376	370	368	352	305	316	361	389	378	360
pH	7	7.1	7.2	7.5	7.7	8.0	8.5	8.2	7.7	7.8	7.9	8
Dissolved Oxygen	6.9	5.8	5.6	6.5	6.6	7.2	10.4	6.6	5.9	5.6	5.2	6.4
Free carbonate	-	8.3	11.8	-	-	-	-	-	-	-	-	9.2
Bi Carbonate Alkalinity	11.7	-	-	14.5	16.8	22.2	17.9	20.7	12.2	13.5	16.6	21.8
Hardness	68.8	69.7	79.8	80.2	92.8	128.9	138.6	108.2	102.6	99.6	90.3	86.7
Calcium	14.3	14.5	14.8	15	15.2	20.7	21.8	26.2	19.7	15.3	14.9	15.2
Magnesium	8	8.1	10.4	10.4	13.3	18.8	20.4	10.4	12.9	14.9	12.6	11.4
Sulphate Sulphur	13.8	6.8	6.3	8.7	16.9	22.3	22.8	26.8	28.6	32	30.8	24
Phosphate-Phosphorus	0543	0.483	0.515	0.520	0.592	0.565	0.573	0.714	0.659	0.812	0.827	0.758
Nitrate Nitrogen	0.513	0.528	0.542	0.552	0.534	0.570	0.619	0.605	0.693	0.676	0.638	0.552
Chloride	54.3	49.8	56.2	59.7	63.8	78.4	82.3	83.6	85.2	113.6	92.4	80.5
BOD	4.64	5.75	5.9	4.12	3.58	3.1	2.5	3.8	3.63	3.92	4	3.94
COD	162	140	142	158	162	169	198	216	236	260	288	262
Dissolved Oxygen matter	6.77	7.22	7.15	5.62	4.53	4.77	2	4.35	6.14	6.79	7.12	9

All values in mg/l, except Temperature (°C), Transparency (cm) conductivity (µmhos/cm) and absent

process alter the fluctuation is drastic in a small water body. The fluctuation of the parameters like free carbonate, Calcium, magnesium, BOD, COD was due to discharge of Organic and inorganic domestic effluents.



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